Power System Protection II

EE 511

Note: In the following slides, there is a slight notation change compared with the previous slides

kr is the receiving end coefficient of reflection

$$v^- = K_r v^+ \qquad \text{where } K_r = \frac{Z_r - Z_0}{Z_r + Z_0} \qquad (8.20)$$

$$i^- = -K_r i^+ \qquad (8.21)$$

Zr is the receiving-end impedance

K_s the coefficient of reflection at the sending-end

$$K_s = \frac{Z_s - Z_0}{Z_s + Z_0} \qquad (8.22)$$

where Z_s is the sending end source impedance.

$$v^+ = Z_0 i^+ \qquad v^- = -Z_0 i^- \qquad (8.17)$$

Z0 is the characteristic impedance of the line

8.3 Superimposed quantities and their properties

$$v_{Rf}(t) = v_{RS}(t) + \Delta v_R(t)$$

$$i_{Rf}(t) = i_{RS}(t) + \Delta i_R(t)$$

The idea of superimposed quantities is best explained by reference to Figure 8.4. Consider point R, where the voltage $v_{Rf}(t)$ and current $i_{Rf}(t)$ are the actual quantities present (see Figure 8.4*a*). These quantities can be split into the sinusoidal prefault quantities $v_{RS}(t)$ and $i_{RS}(t)$ shown in Figure 8.4*b*, and the fault superimposed quantities $\Delta v_R(t)$ and $\Delta i_R(t)$ shown in Figure 8.4*c*. Thus

$$\begin{aligned} v_{Rf}(t) &= v_{RS}(t) + \Delta v_R(t) \\ i_{Rf}(t) &= i_{RS}(t) + \Delta i_R(t) \end{aligned} \tag{8.34}$$

It should be noted that the sum of the steady-state and superimposed quantities is equal to the total variation at all points in the network. The superimposed network is therefore a network with all source voltages set at zero (short circuited). The superimposed fault point voltage is therefore zero for all time up to the time of the fault disturbance. The voltage of the fictitious superimposed source on the faulted phase(s) is thus equal in magnitude and opposite in sign to the prefault voltage at the fault point (i.e. $v_F(t)$).

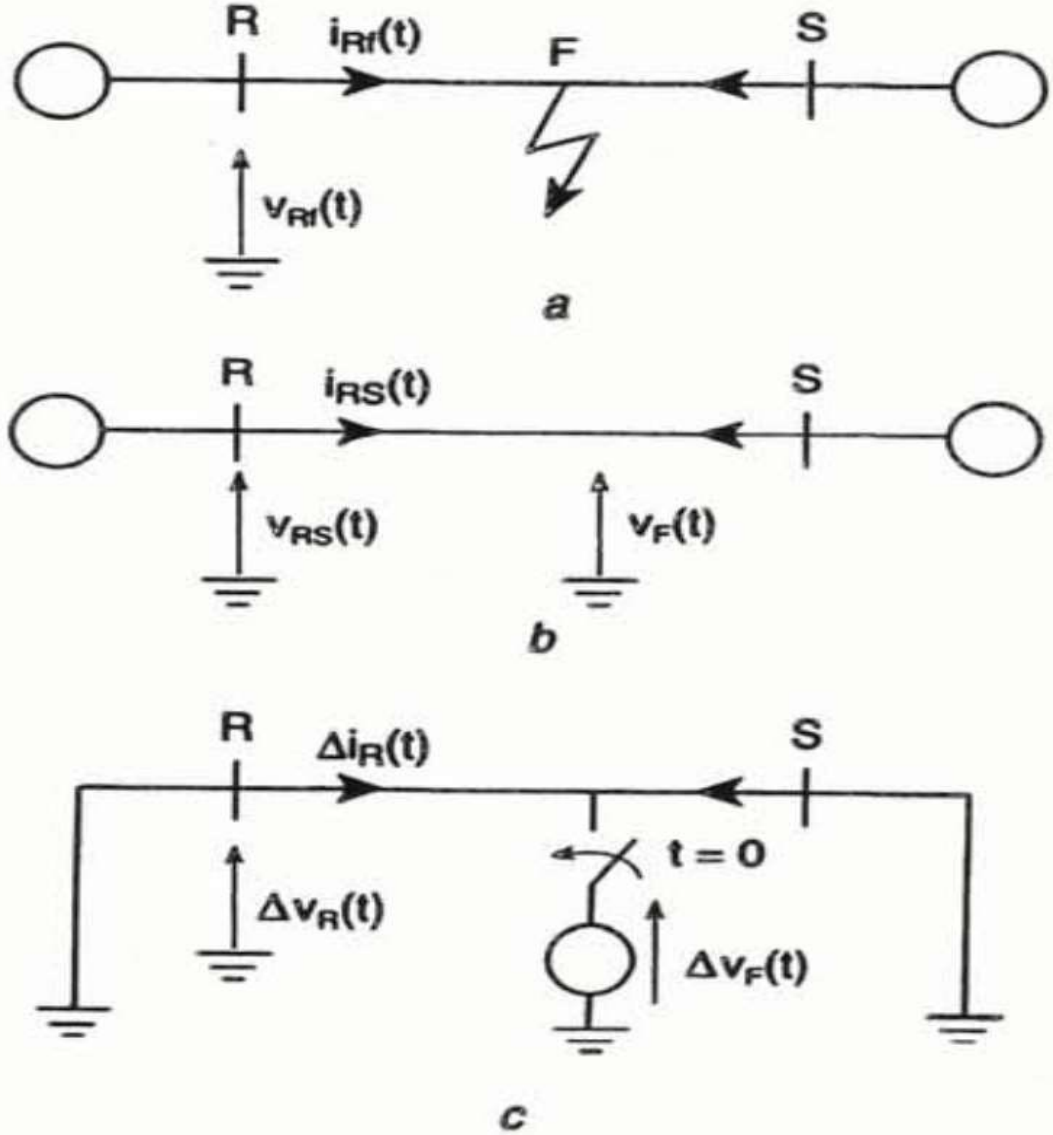
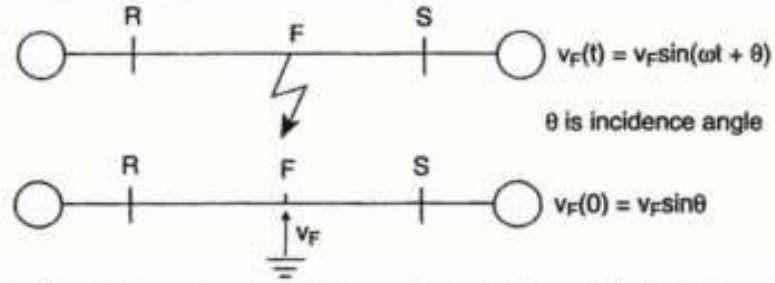


Figure 8.4 *Superimposed voltage and current generated by a fault*

(*a*) actual system under a fault condition
(*b*) the steady-state (prefault) network
(*c*) the superimposed network

8.3.1 Polarity of superimposed quantities versus fault location

If an internal fault is assumed to occur during the positive half cycle of the prefault voltage waveform, then the signs of the superimposed voltage and current would be (−) and (+), respectively, at both ends of the line. (Note that reference directions are those shown in Figure 8.4). However, if the fault occurs during the negative half cycle of the voltage waveform, the polarities of the superimposed voltage and current at the R and S ends would be (+) and (−) respectively. Repeating the analysis for faults at locations external to the line R–S reveals that the polarities of the superimposed voltage and current would be similar to each other at one end but are different from each other at the other end. Figure 8.5 explains in detail all possible cases.



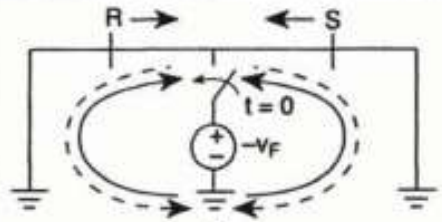
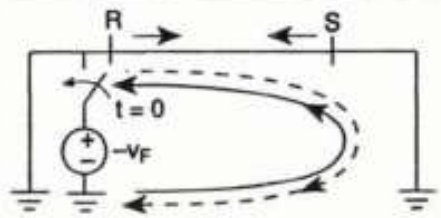
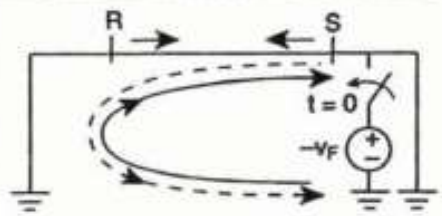
type of fault	corresponding superimposed network	v_F polarity	polarity of superimposed components			
			Δi_R	Δv_R	Δi_S	Δv_S
internal fault	 *a*	pos.	+	−	+	−
		neg.	−	+	−	+
external fault beyond bus R	 *b*	pos.	−	−	+	−
		neg.	+	+	−	+
external fault beyond bus S	 *c*	pos.	+	−	−	−
		neg.	−	+	+	+

Figure 8.5 Dependency of polarities of superimposed voltage and current quantities on faul location

(i) the line is lossless, and therefore the surge impedance of the line (Z_0) and the wave propagation velocity c are given by

$$Z_0 = \sqrt{L/C}$$
$$c = 1/\sqrt{LC} \approx 300 \times 10^8 \text{ m/s} \tag{8.35}$$

(ii) the surge impedances looking into systems S and R are equal to Z_{0S} and Z_{0R}, respectively, and are such that

$$Z_{0R} = Z_{0S} = Z_0/3 \tag{8.36}$$

Reverse faults

Let us assume that a solid fault occurs immediately behind the relay location at R. Now because the fault point coincides with the relaying point, the resulting superimposed voltage at R, $\Delta v_R(t)$, is equal and opposite to the steady-state prefault voltage waveform $v_F(t)$ at end R, i.e. $\Delta v_R(t) = -v_F(t)$.

this causes a travelling wave to propagate towards end S at a velocity c defined by eqn. 8.35. The associated superimposed current waveform $\Delta i_R(t)$ also propagates at a velocity c and the superimposed voltage and current

waveform $\Delta v_R(t)$ and $\Delta i_R(t)$ are interrelated according to eqn. 8.17. In the period of time from fault inception up to twice the wave transit time from end R to S ($2l/c$, l = length of line), $\Delta v_R(t) = \Delta v_R^+(t)$, $\Delta i_R(t) = \Delta i_R^+(t)$ and $\Delta v_R^-(t) =$

$\Delta i_R^-(t) = 0$; it therefore follows that, during this period, the superimposed voltage and current at the relaying point R are related by

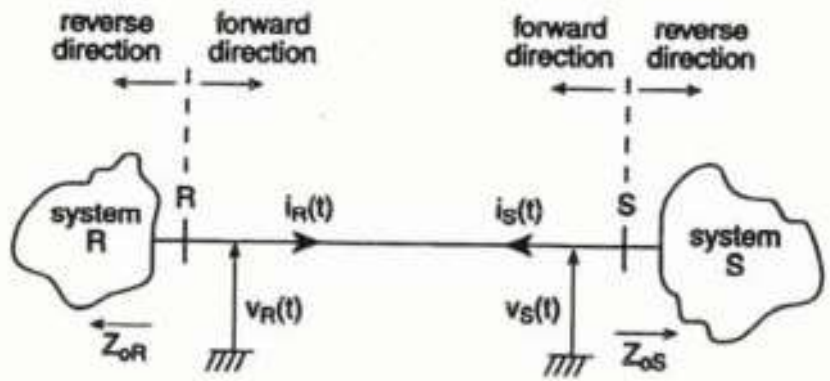


Figure 8.6 Line interconnecting two systems R and S

$$\left.\begin{aligned}\Delta v_R(t) &= Z_0 \Delta i_R(t)\\ \Delta v_R(t) - Z_0 \Delta i_R(t) &= 0\end{aligned}\right\} \tag{8.37}$$

or

Because S is a point of discontinuity, part of the incident superimposed voltage and current waveforms are reflected back towards the relaying point at R, according to eqns. 8.20 and 8.21, such that

$$\begin{aligned}\Delta v_S^-(t) &= K_r \Delta v_S^+(t)\\ \Delta i_S^-(t) &= -K_r \Delta i_S^+(t)\end{aligned} \tag{8.38}$$

where

$$K_r = (Z_{0S} - Z_0)/(Z_{0S} + Z_0)$$

and $\Delta v_S^+(t)$, $\Delta i_S^+(t)$ are the transmitted parts of superimposed voltage and current waveforms, respectively, at end S, and $\Delta v_S^-(t)$, $\Delta i_S^-(t)$ are the reflected parts of the superimposed voltage and current waveforms, respectively, at end S.

It will be evident that it takes a time of $2\ l/c$ for the waveform reflected from end S to arrive back at R after the occurrence of the fault. Figure 8.7 thus illustrates the behaviour of the system shown in Figure 8.6 due to a fault that occurs at the positive-to-negative zero crossing of prefault voltage. Figure 8.7(*b*) shows the lattice diagram of the superimposed voltages and currents at any point on the line, and Figure 8.7(*c*) shows the resulting superimposed components $\Delta v_R(t)$, $\Delta i_R(t)$ at the relaying point R.

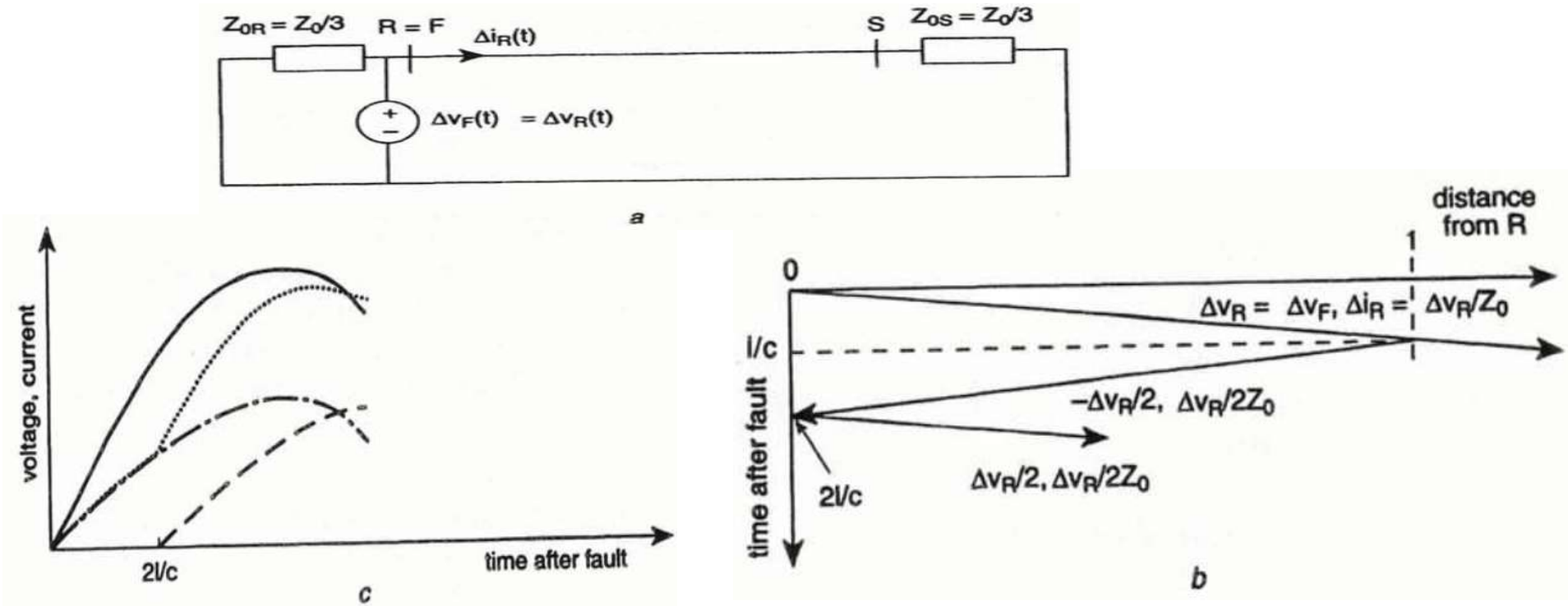


Figure 8.7 *Initial propagation of superimposed quantities following a reverse fault immediately behind the relay at R*

(*a*) superimposed fault network
(*b*) lattice diagram
(*c*) time of variation of superimposed quantities consistent with fault at the positive-to-negative zero-crossing of the prefault voltage

——— total superimposed voltage at R
$\Delta v_R(t) = \Delta v_F(t) = h(t) V_m \sin \omega_0 t$

—·— initial superimposed current $\Delta v_F(t)/Z_0$

– – – – total reflected superimposed current in line at R at time $2l/c$, $\Delta v_F(t - 2l/c)/Z_0$

------ total superimposed current in line at R
$\Delta i_R(t) = \Delta v_F(t)/Z_0 + \Delta v_F(t - 2l/c)/Z_0$

Forward faults

Now let us consider the case where faults occur in the forward direction with respect to the relay at end R. For example, consider a fault at a point F which is at a distance x_F from R. The resulting superimposed voltage at x_F, $\Delta v_F(t)$, is equal and opposite to the prefault voltage at that point. Again in accordance with Section 8.2.1, the application of the superimposed voltage $\Delta v_F(t)$ causes the voltage travelling wave and its associated current wave $\Delta v_F(t)/Z_0$ to propagate at a velocity c towards the point R. The superimposed voltage and superimposed current waveforms incident at R arrive at a time x_F/c and are given by

$$\Delta v_R(t) = \Delta v_R^+(t) = \Delta v_F(t - x_F/c)$$

$$\Delta i_R(t) = -\Delta i_R^+(t) = -\Delta v_F(t - x_F/c)/Z_0$$

It is notable that the negative sign in

$$\Delta i_R(t) = -\Delta i_R^+(t)$$

is due to the fact that in Fig.8.7(a) the positive direction of the current is defined from left to the right. However, $\Delta i_R^+(t)$ flows from right to left.

Now the reflected components of these voltage and current waveforms for an arbitrary terminating surge impedance Z_{0R} can be found from the corresponding incident waves and the reflection coefficient:

$$\Delta v_R^-(t) = K_r \Delta v_R^+(t)$$

$$= \Delta v_F(t - x_F/c)\,(Z_{0R} - Z_0)/(Z_{0R} + Z_0) \tag{8.39}$$

$$\Delta i_R^-(t) = -K_r \Delta i_R^+(t)$$

$$= -\Delta v_F(t - x_F/c)\,(Z_{0R} - Z_0)/[(Z_{0R} + Z_0)Z_0] \tag{8.40}$$

But the total variation of superimposed voltage and current at, and immediately after, the time when the waves reach end R ($t = x_F/c$) is given by the sum of transmitted and reflected waves:

$$\Delta v_R(t) = \Delta v_R^+(t) + \Delta v_R^-(t)$$

$$= 2Z_{0R}\Delta v_F(t - x_F/c)/(Z_{0R} + Z_0) \tag{8.41}$$

$$\Delta i_R(t) = \Delta i_R^+(t) + \Delta i_R^-(t)$$

$$= -2\Delta v_F(t - x_F/c)/(Z_{0R} + Z_0) \tag{8.42}$$

Therefore, by finding the difference between eqns. 8.41 and 8.42 and their sum, we obtain

$$\Delta v_R(t) - Z_0\Delta i_R(t) = 2\Delta v_F(t - x_F/c)$$

and

$$\Delta v_R(t) + Z_0\Delta i_R(t) = 2\Delta v_F(t - x_F/c)\,(Z_{0R} - Z_0)/(Z_{0R} + Z_0) \tag{8.43}$$

Eqns. 8.43 show that, unlike the previously considered reverse fault, the difference $\Delta v_R(t) - Z_0\Delta i_R(t)$ becomes immediately finite on arrival of the superimposed waves at x_F/c after fault inception. This conclusion holds for faults anywhere on the line or within system S.

Figure 8.8 illustrates the behaviour of the superimposed quantities following a fault on the line R–S of the system of Figure 8.6. The fault occurs at the negative peak of the prefault voltage. It is evident from the lattice diagram and the waveforms of Figures 8.8(*b*) and (*c*), that, initially, the travelling waves of superimposed voltage and current are of opposite polarity.

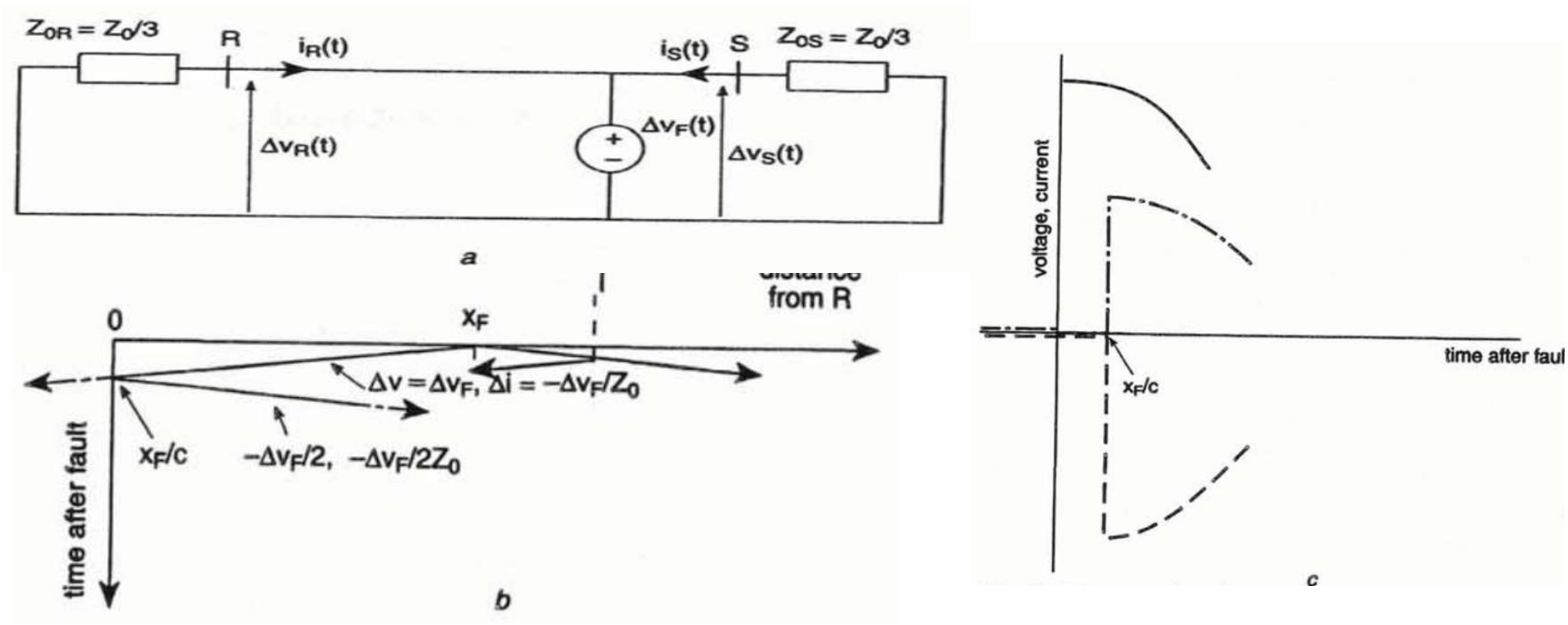


Figure 8.8 *Initial propagation of superimposed quantities following a fault in the forward direction of the relay at R*

(*a*) superimposed fault network
(*b*) lattice diagram
(*c*) time variation of superimposed quantities consistent with fault at negative peak of prefault voltage

——— superimposed fault voltage $v_F(t)$
—·— superimposed voltage at R
$v_R(t) = \Delta v_F(t - x_F/c)/2$
------ superimposed current at R
$i_R(t) = -3\Delta v_F(t - x_F/c)/2Z_0$

Relaying signals

The difference in the behaviour of the superimposed quantities according to whether the fault is in the reverse or the forward direction can be used to determine the direction of the fault with respect to the measuring point. This is done by the formation of two signals. In terms of primary system values, the signals used at R are

$$S_{1R} = \Delta v_R(t) - R_0 \Delta i_R(t)$$
$$S_{2R} = \Delta v_R(t) + R_0 \Delta i_R(t) \tag{8.44}$$

The corresponding signals used at S which have the reference directions defined in Figure 8.8(*a*), are given as

$$S_{1S} = \Delta v_S(t) - R_0 \Delta i_S(t)$$
$$S_{2S} = \Delta v_S(t) + R_0 \Delta i_S(t) \tag{8.45}$$

where R_0 is a surge replica resistance whose value is so arranged as to match closely the line surge impedance Z_0.

By examining the signals at end R for reverse faults, with the assumption of lossless line and perfect matching of the replica resistance R_0, i.e. by substituting eqn. 8.37 into eqns. 8.44, we obtain

$$S_{1R} = 0$$
$$S_{2R} = 2\Delta v_R(t) \tag{8.46}$$

Figure 8.9(*a*) shows how the signals vary following the reverse fault condition considered previously. It is important to note that S_{2R} becomes finite before S_{1R}. Figure 8.9(*a*) also shows the sequence in which the signals S_{1R} and S_{2R} rise above a given threshold value $|V_s|$, (S_{2R} followed by S_{1R}), at times $T_1 < T_2$.

Similarly, the signals S_{1R} and S_{2R} for the forward fault described previously can be found by substituting eqn. 8.43 into eqn. 8.44. By assuming perfect matching of the replica resistance R_0 to the surge impedance Z_0, these signals would be given as

$$S_{1R} = 2\Delta v_F(t - x_F/c)$$
$$S_{2R} = 2\Delta v_F(t - x_F/c)\,(Z_{0R} - Z_0)/(Z_{0R} + Z_0) \tag{8.47}$$

Figure 8.9(*b*) illustrates the behaviour of the signals S_{1R}, S_{2R} due to the forward fault condition on the system of Figure 8.8. It can be seen that the setting levels V_s and $-V_s$ are exceeded almost simultaneously ($T_1 \approx T_2$) owing to the very rapid change in the superimposed components associated with the assumed fault at peak prefault voltage.

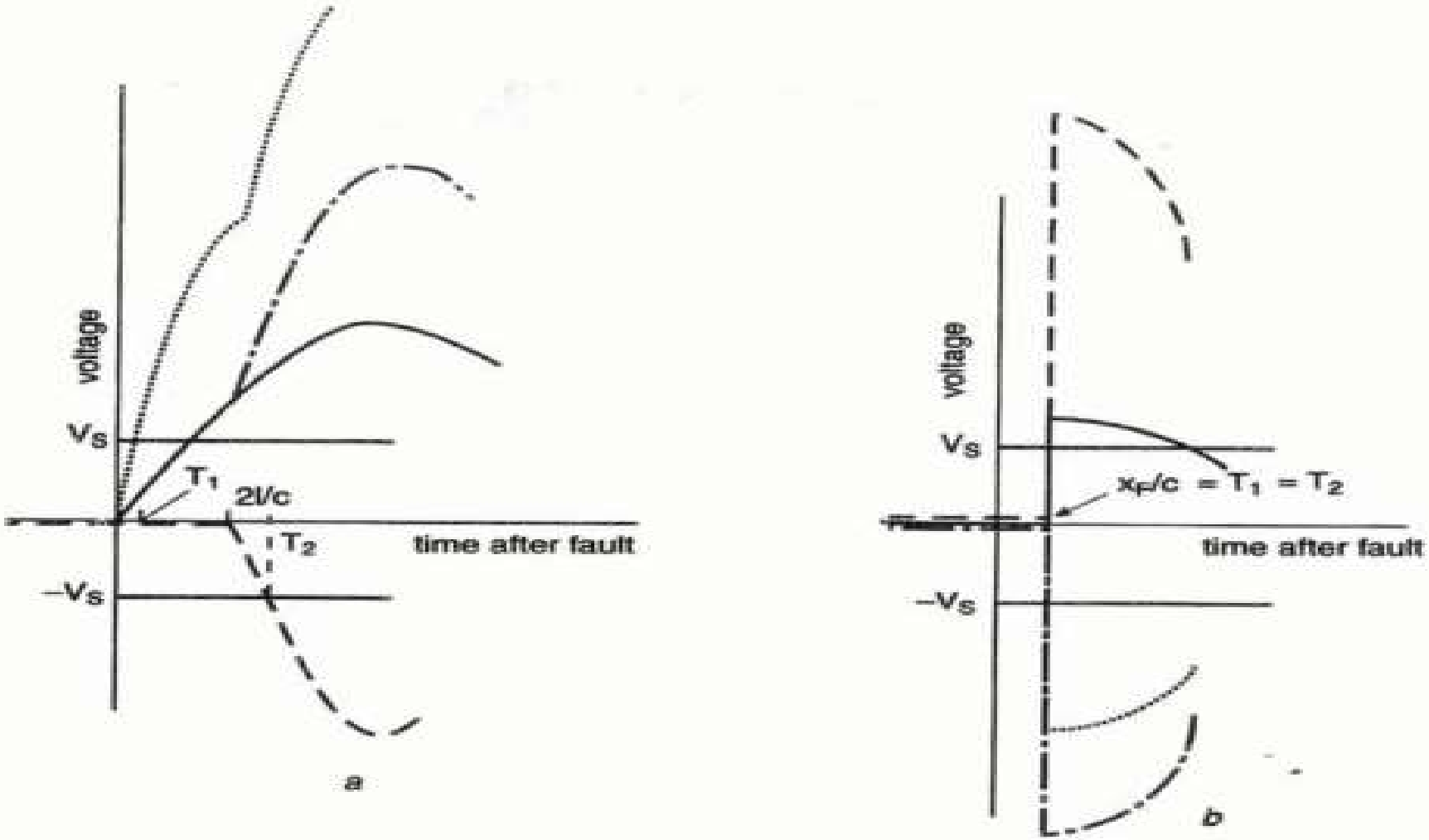


Figure 8.9 *Variation of relaying signals at R following faults on the system of Figure 8.6*

(*a*) reverse fault at R
(*b*) fault on line R–S

——— $\Delta v_R(t)$
—·—·— $R_0 \Delta i_R(t)$
— — — $S_{1R} = \Delta v_R(t) - R_0 \Delta i_R(t)$
------ $S_{2R} = \Delta v_R(t) + R_0 \Delta i_R(t)$

When a three-phase line is considered, the phase variations of currents and voltages are decomposed into modal variations, and the relaying signals can therefore be expressed in terms of aerial-mode surge impedances $Z_0^{(1)}$ and $Z_0^{(2)}$, and the earth-mode surge impedance $Z_0^{(0)}$ (see Section 8.2.2). Assuming a lossless three-phase line with complete transposition and perfect matching of the replica surge impedances $R_0^{(k)}$, $k=0$, 1 and 2 to the corresponding surge impedance $Z_0^{(k)}$, $k=0$, 1 and 2, the modal relaying signals at the end R of the line are given by

$$\begin{aligned} S_{1R}^{(k)} &= \Delta v_R^{(k)}(t) - Z_0^{(k)} \Delta i_R^{(k)}(t) \\ S_{2R}^{(k)} &= \Delta v_R^{(k)}(t) + Z_0^{(k)} \Delta i_R^{(k)}(t) \end{aligned} \qquad k=0,\ 1,\ 2 \tag{8.48}$$